

Zheng Fang

Able to Relocate | fangzheng.apply@gmail.com | 740-405-4906 | [LinkedIn](#) | [GitHub](#)

EDUCATION

Washington University in St. Louis

St. Louis, MO

Master in Engineering Data Analytics & Statistics

Aug. 2025 - May 2026

Bachelor of Science, double major in Data Science and Financial Engineering

Aug. 2023 - May 2025

Cumulative GPA: 3.78/4.00 | Engineering GPA: 3.95/4.00

Honors: Dean's List of All Semester, Lee Hunter Scholarship, Graduate Affiliation Scholarship

Denison University

Granville, OH

Bachelor of Arts in Mathematics

Aug. 2020 - May 2023

Honor: Denison Scholarship, Denison Grant, Introductory Physics (Book) Award, VanBeuren Scholarship

SKILLS

Programming Languages: Python (Pandas, NumPy, Scikit-learn), C++, Julia, R, Java, SQL (MySQL), Oracle

Data Visualization: Tableau, Python (Matplotlib, Seaborn, Plotly), Excel, Power BI

Machine Learning Techniques: Linear Regression, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, SVM, KNN, Neural Networks, Clustering, PCA, XGBoost, Regularization, Feature Engineering, Model Evaluation, CNN, RNN (LSTM)

Statistical Analytics: Hypothesis Test, T-test, Chi-square Test, Exploratory Data Analysis, A/B Testing, ANOVA, Distributions

WORK EXPERIENCE

ByteDance

Beijing, China

Data Science Intern (Data Pre-training), Tools: Python, A/B Test

May 2024 - Aug. 2024

- Engineered Python-based data dashboards to monitor pre-training metrics for Large Language Models (LLMs) using **Matplotlib** and **Seaborn**, and conducted **A/B testing** to refine data annotation processes, providing enhancement proposals to the R&D team
- Crafted prompts for large models to facilitate structured pre-training data annotation, developed workflows integrating tasks between human input and programmed execution, enhancing 5% pre-training efficacy and maintaining quality
- Refined prompts across multiple revisions, iterating 10+ fine-tuned versions tailored to project-specific requirements
- Managed annotation projects aligning with R&D objectives, designed daily operational standards, and maintained data quality

Sinochem Corporation

Beijing, China

Data Engineer Intern, Tools: SQL, HIVE, ETL

May 2023 - Aug. 2023

- Leveraged **SQL** and **HIVE** to conduct statistical analysis on large datasets, including data preprocessing, aggregation, and mining
- Selected appropriate statistical analysis methods based on business problems and data characteristics
- Managed daily maintenance of the **data warehouse**, designed and executed **ETL** processes for data extraction, transformation, and loading, and monitored data flow and updates, promptly identifying and resolving data transmission issues
- Designed and built data warehouses for HSE safety and risk monitoring platforms, conducted real-time monitoring of business data, assessed and scored risks based on risk maps, and promptly reported abnormal data to upstream and downstream stakeholders

TECHNICAL PROJECTS

Housing Price Prediction Using Machine Learning (Python, Scikit-learn, PyTorch, XGBoost)

Dec. 2024

- Competed in a Kaggle challenge to predict housing prices above or below \$200,000, achieving **92% accuracy** on a private leaderboard
- Preprocessed data with advanced feature engineering, such as scaling, encoding categorical variables, and handling missing data
- Implemented and compared multiple machine learning models, such as **Logistic Regression**, **Random Forest**, **XGBoost**, and **Neural Networks (PyTorch)**, to identify the best-performing approach
- Fine-tuned submission format, predicting binary outcomes (0/1) for a test dataset and submitting valid ID-200K mappings
- Conducted rigorous error analysis, visualizing training and validation losses over **5000 iterations** to refine model performance

Job Change of Data Scientists (Python, EDA)

Apr. 2023 - Jun. 2023

- Predicted job change willingness using supervised machine learning models (**random forest**, **XGBoost**, and **KNN**)
- Performed data visualization with **Matplotlib** and **Seaborn** on feature correlation to identify needs for preprocessing
- Applied data preprocessing procedures, including missing value imputation, feature encoding, and normalization by **Scikit-learn**
- Improved the XGBoost model by tuning the maxim of depth to 15, achieving a 5% increase in the recall on prediction of minority job change intentions
- Implemented cross-validation to avoid overfitting and used confusion matrix to evaluate model performance

Sentimental Analysis on New York Time (Python)

Apr. 2023 - Jun. 2023

- Sanitized and collected 1000+ articles data through New York Time **Python API** requests and extracted **JSON** response
- Segmented JSON news content from the response and applied Elastic Search to change Lucene Query and retrieve articles
- Utilized the **NLTK** library for NLP-based tokenization by removing stop words and stemming meaningful words
- Improved **VADER** sentiment analysis model by customizing lexicon with COVID-related tokens to reflect the public emotion changes

Uber Marketing Analysis (Python)

Feb. 2023 - Mar. 2023

- Performed **SWOT analysis** to assess competitive landscape of Uber, Didi Chuxing, and Lyft, focusing on price, mileage, and user experience, and developed strategic plans
- Applied **clustering algorithms** from **sklearn** to segment user behavior, contributing to understand user demographics and facilitated improvement in user retention and satisfaction
- Analyzed Uber's research report data to address low driver retention and losses, evaluated Uber's business model, expansion strategies, and customer engagement strategies to develop sustainable profitability improvement